

Checklist of Learning Objectives

MAT SCI 120 is a course that is quite hard to master because of its focus on physics. Try to stay up-to-date with everything that is described in the lectures, discussions, and homeworks when studying for the exams. You can also try to re-derive everything for the course to make sure you understand the basic objectives of the course. To facilitate the process of studying, this checklist provides the essential learning objectives for the course. It is not a comprehensive list of everything in the course, but it shows reasonable coverage of the coursework and can help you verify if you are following the course during your studies. If you know everything on this checklist, you should be ready for any exam. When studying for exams or reviewing the checklist below, try to ask yourself: “do I understand how to ...?” This will guide your learning process and help you identify knowledge gaps. The checklist is organized by area of knowledge in our course (see syllabus).

Overall Learning Objectives

Physics of Materials (MAT SCI 120)

1. Describe the interplay between wave and particle behavior of electrons and their influence in explaining solid-state property of materials
2. Interpret the postulates of quantum mechanics, solve simple cases of the Schrödinger's equation, and describe the solutions for the electron in the hydrogen atom.
3. Explain how atoms and electrons behave in a lattice, including approximations such as the free electron model, phonons, and the origin of the band gap and effective masses of electrons/holes.
4. Interpret electronic band structures, electronic density of states, and band diagrams for several junctions, including semiconductor materials.
5. Quantitatively describe the operation mechanisms of semiconductor devices, from P-N junctions to MOSFETs, and what is the physical origin of their behavior.
6. Quantitatively describe the physical origins of optical, dielectric, and magnetic properties of materials.

Pre-requisites (MAT SCI 104)

- Understand the periodic table and elemental properties such as electronegativity
- Write down the electronic configuration of neutral and charged elements.
- Explain the origin of chemical bonds based on electronic energy states
- Classify between different bonding types (ionic, covalent, metallic, secondary)
- Broadly explain trends in bonding type and expected properties (e.g., conductivity, density)
- Explain what are crystals and recognize the four main structure types (SC, FCC, BCC, HCP) and their coordination numbers
- Recognize crystallographic directions in a crystal structure and their associated planes.
- Remember qualitatively the electronic and optical properties of materials studied in MAT SCI

104.

- Remember what are the defect types and their dimensionality.
- Understand Fick's first and second law of diffusion, and how they are related to temperature.

Pre-requisites (MAT SCI 110)

- Understand what is a reciprocal lattice and how to compute reciprocal vectors
- Understand the relationship between diffraction patterns and crystal lattices
- Explain how interference is used to perform measurements in crystal structures, and the interpretation of the atomic form factor.

Pre-requisites (Physics)

- Remember about basic electrostatics and electromagnetism, including Lorentz's force, the definition of an electric field, conservative field, and electrical potential energy.
- Recognize the Maxwell's equations with their local and integral forms, and understand how to solve simple electrostatics and electrodynamics problems given boundary conditions.
- Understand the mathematical formalism related to the wave equation.
- Discuss the features of Bohr's model for the hydrogen atom, and the process leading to the quantization of the angular momentum of the electron.

Pre-requisites (Math)

- Understand and solve problems related to vector calculus and multi-variate calculus (including gradient, divergence, curl, and Laplacian)
- Understand how to solve ordinary differential equations and partial differential equations, with a specific focus on understanding the homogeneous solutions and boundary conditions
- Transform between systems of coordinates, specifically within cartesian, cylindrical, and spherical coordinates, and how that modifies the solutions to vector calculus
- Understand, interpret, and plot mathematical equations, explaining their limits and behavior.
- Understand the concept of probability distributions, random variables, expected values, and density functions.

Electron as a particle (ch. 01)

- Derive the conductivity formula $\sigma = \frac{N_e e^2 \tau}{m} = \mu_e N_e e$ from drift velocity and collision time concepts, explaining how it validates Ohm's law.
- Understand how to compute the Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Explain how the Hall effect measurement with $E_H = R_H J B$ and $R_H = \frac{1}{N_e e}$ reveals carrier type through sign reversal and enables experimental determination of carrier density.

Electromagnetic waves (ch. 01)

- Understand the physical meaning of each of the Maxwell's equations:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \quad (1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (4)$$

- Know how to solve simple problems with given boundary conditions with Maxwell's equations
- Calculate the plasma frequency $\omega_p = \sqrt{\frac{N_e e^2}{m \epsilon_0}}$ and interpret its role as the threshold separating opaque and transparent regimes in metals.
- Apply the skin depth formula $\delta = \sqrt{\frac{2}{\omega \mu \sigma}}$ to determine the limits of electromagnetic wave propagation in conductors and explain why skin depth decreases with increasing frequency.
- Derive the cyclotron resonance condition $\omega = \omega_c = \frac{e B_0}{m}$ and explain how it enables direct measurement of effective electron mass when $\omega_c \tau \gg 1$.

Electron as a wave (ch. 02)

- Derive the de Broglie wavelength $\lambda = \frac{h}{mv}$ and explain how it connects particle momentum to wave properties.
- Distinguish between wave interference and particle addition as something that separates quantum and classical behavior.
- Apply the Bragg diffraction condition $n\lambda = d \sin \theta$ to measure electron wavelengths in crystal scattering experiments such as Davisson-Germer.
- Explain why Heisenberg's uncertainty principle $\Delta p \cdot \Delta z \geq \hbar/2$ represents a fundamental property of quantum reality rather than a limitation of a measurement apparatus.
- Distinguish between group velocity $v_g = \partial \omega / \partial k$ and phase velocity $v_p = \omega / k$, identifying group velocity as the electron's actual velocity.
- Relate the energy-time uncertainty principle to the momentum-position uncertainty principle as analogous fundamental limits.
- Connect wave-particle duality concepts to their foundational role in understanding electron behavior in solids.

Quantum mechanical behavior of electrons (ch. 03)

- Explain the probabilistic interpretation of the wave function ψ and why $|\psi|^2$ represents probability density while ψ itself is not directly observable.

- Explain why the Schrödinger equation is a fundamental postulate of quantum mechanics, validated by experimental agreement rather than derived from first principles.
- Derive the time-independent Schrödinger equation $-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = E\psi$ from the time-dependent form using separation of variables.
- Connect the momentum operator $\hat{p} = -i\hbar\nabla$ to the de Broglie relation $p = \hbar k$, recovering the classical wavelength-momentum relationship.
- Apply boundary conditions (continuity of ψ and $\partial\psi/\partial z$) at potential discontinuities to determine reflection and transmission coefficients.
- Explain quantum tunneling as exponential decay of the wave function in classically forbidden regions where $E < V$, with $\psi_2 \propto e^{-|k_2|z}$.
- Calculate energy levels in a rigid potential well using $E_n = \frac{h^2 n^2}{8mL^2}$ and explain why the ground state has non-zero energy (zero-point energy).
- Interpret the transcendental equation $k \tan\left(\frac{kL}{2}\right) = \gamma$ for finite potential wells and explain why graphical solutions yield a finite number of bound states.
- Apply the uncertainty relations $\Delta x \Delta p \geq \hbar$ and $\Delta E \Delta t \geq \hbar$ to explain measurement limits and energy borrowing in tunneling processes.
- Calculate expectation values using $\langle A \rangle = \frac{\int_V \psi^* \hat{A} \psi dV}{\int_V |\psi|^2 dV}$ to obtain average position, momentum, and energy from wave functions.
- Compare finite and infinite potential wells, distinguishing why finite wells support a maximum number of bound states while infinite wells have unlimited discrete levels.

Hydrogen atom (ch. 04)

- Describe how one can derive the ground state energy of hydrogen $E_1 = -\frac{me^4}{8\epsilon_0^2\hbar^2} = -13.6 \text{ eV}$ from the time-independent Schrödinger equation with Coulomb potential $V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$.
- Explain why the ground state wave function $\psi \propto e^{-c_0 r}$ has exponential rather than oscillating form and how the Bohr radius $a_0 = 0.0528 \text{ nm}$ represents the maximum probability location rather than a classical orbit.
- Identify the four quantum numbers (n, ℓ, m_ℓ, s) that completely specify an electron state and explain why hydrogen energy levels $E_n = -13.6/n^2 \text{ eV}$ depend only on the principal quantum number n .
- Calculate approximate energy levels for multi-electron atoms using the independent electron model $E_n = -13.6 \frac{Z_{\text{eff}}^2}{n^2} \text{ eV}$ and identify its limitations.
- Connect the large energy jumps between shells to the periodic structure of chemical properties.
- Interpret radial probability density plots $r^2|\psi|^2$ for 1s, 2s, and 2p states, identifying nodes and their physical significance for shell structure.

- Distinguish between the spherically symmetric Laplacian $\nabla^2\psi = \frac{\partial^2\psi}{\partial r^2} + \frac{2}{r}\frac{\partial\psi}{\partial r}$ and its cartesian counterpart, explaining its role in solving the hydrogen atom versus general three-dimensional problems.

Chemical bonds (ch. 05)

- Explain why chemical bonds form through energy minimization, identifying how attractive and repulsive forces balance at equilibrium separation r_0 to create a stable configuration.
- Distinguish among the four bond types—ionic, metallic, covalent, and van der Waals—and predict how each determines material properties including hardness, electrical conductivity, ductility, and melting point.
- Interpret why quantum coupling between two states always produces energy level splitting into bonding ($E_0 - A$) and antibonding ($E_0 + A$) states, with the lower state providing stabilization.
- Using the square double well example, describe why electron localization between atomic sites decreases the energy of the system and explains covalent bond formation in systems like H_2^+ .
- Explain the physical criterion for when repulsive force overcomes attractive force in chemical bonds

Free electron theory (ch. 06)

- Explain why the infinite potential well approximation is valid for modeling valence electrons in metals despite the finite work function, and identify the conditions under which this simplification breaks down.
- Derive the density of states $Z(E) = CE^{1/2}$ from three-dimensional phase space arguments, where $C = \frac{4\pi L^3(2m)^{3/2}}{h^3}$, and explain why energy levels become quasi-continuous at macroscopic scales.
- Calculate the Fermi energy from electron density N/V using $E_F = \frac{\hbar^2}{2m}(3\pi N/V)^{2/3}$ and explain why this quantity is independent of temperature.
- Apply the Fermi-Dirac distribution $F(E) = \frac{1}{\exp\{(E-E_F)/k_B T\}+1}$ to determine electron occupation probabilities and explain why the Pauli exclusion principle gives rise to this distribution. What would happen if the Pauli exclusion principle did not exist?
- Distinguish between thermionic emission, field emission, and photoelectric emission. Identify that field emission occurs via quantum tunneling and is temperature-independent while thermionic emission depends exponentially on temperature.
- Interpret Einstein's photoelectric effect explanation using the threshold condition $\hbar\omega > \varphi$ and connect this to the work function of a material.

Band theory of solids (ch. 07)

- Explain why the free-electron model fails to distinguish metals from insulators and how band theory resolves this by introducing allowed energy bands separated by forbidden gaps.

- Understand the Bloch wave function form $\psi_k(x) = u_k(x)e^{ikx}$ and explain why electrons in periodic potentials satisfy this structure.
- Identify that energy discontinuities (band gaps) occur at wave vectors $k = n\pi/a$ due to Bragg reflection, which creates standing waves and splits energy bands.
- Calculate effective mass using $m^* = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2} \right)^{-1}$ and interpret how band curvature determines electron response to applied forces.
- Interpret negative effective masses near band edges as hole-like behavior where electrons accelerate opposite to the applied force direction.
- Explain why completely filled bands contribute to zero conduction electrons. Explain this by referring to the group velocity $v_g = \frac{1}{\hbar} \frac{\partial E}{\partial k}$ vanishing at band edges.
- Distinguish between metals (partially filled bands), insulators (completely filled bands with large gaps), and semiconductors (filled bands with smaller gaps) using band structure diagrams and Fermi level position.
- Explain the Kronig-Penney model, explaining how a band structure emerges through quantum mechanical behavior of electrons in a periodic potential
- Explain how thermal energy at finite temperatures promotes electrons across band gaps, creating electron-hole pairs and converting insulators into semiconductors.
- Identify holes as mathematical constructs representing missing electrons in nearly full bands, possessing positive charge and effective mass opposite to the missing electron.

Semiconductors (ch. 08)

- Explain why intrinsic semiconductor conductivity follows $N_{e,h} \propto \exp(-E_g/2k_B T)$ while metals show decreasing conductivity with temperature.
- Explain the temperature dependence of conductivity in intrinsic and extrinsic semiconductors. Describe how dopant concentration changes the mobility and carrier concentration in intrinsic semiconductors.
- Derive the electron concentration in the conduction band using $N_e = N_c \exp\left(-\frac{E_g - E_F}{k_B T}\right)$ and identify when the Boltzmann approximation is valid.
- Apply the charge neutrality condition $N_e + N_A^- = N_h + N_D^+$ to determine Fermi level position in doped semiconductors.
- Interpret the mass action law $N_e N_h = N_i^2$ and explain why it holds regardless of doping level.
- Distinguish between direct-gap semiconductors (GaAs, InP) and indirect-gap semiconductors (Si, Ge) in terms of electron excitation (photon only vs. photon + lattice scattering).
- Calculate effective mass from cyclotron resonance measurements using $m^* = \frac{eB}{\omega_c}$.

- Explain how the effective mass dispersion relation $E = \frac{\hbar^2}{2m^*}(k_x^2 + k_y^2 + k_z^2)$ simplifies carrier transport analysis.
- Apply the Fermi level expression $E_F = \frac{E_g}{2} + \frac{3}{4}k_B T \log_e \left(\frac{m_h^*}{m_e^*} \right)$ to predict Fermi level position in intrinsic semiconductors.

Semiconductor devices (ch. 09)

- Explain why contact between p-type and n-type semiconductors forces Fermi level equilibration, connecting E_F to both chemical potential (thermodynamics) and carrier concentration (kinetics via Fick's laws).
- Derive the charge density, electric field, and electrostatic potential in the depletion region using the Poisson equation under the depletion approximation, and sketch $\rho(x)$, $E(x)$, $V(x)$.
- Derive the built-in voltage both electrostatically, $V_{bi} = \frac{e}{2\epsilon}(N_A x_p^2 + N_D x_n^2)$, and thermodynamically, $V_{bi} = \frac{k_B T}{e} \ln \left(\frac{N_A N_D}{n_i^2} \right)$, and explain why equating them determines the depletion widths.
- Explain what is the depletion zone and whether carriers are moving or are static in the depletion zone.
- Apply charge neutrality ($N_A x_p = N_D x_n$) to determine which side of an asymmetric junction hosts the wider depletion region and derive the total depletion width W .
- Describe how forward bias ($U_1 > 0$) narrows the depletion region and lowers the barrier, while reverse bias ($U_1 < 0$) widens it, and identify the strong forward bias regime where ohmic conduction dominates.
- Derive the Shockley diode equation $I = I_0 [\exp(eU_1/k_B T) - 1]$ from Boltzmann statistics, explaining why diffusion current is bias-dependent while drift current is approximately bias-independent.
- Identify the limiting behaviors of the diode equation: exponential growth in forward bias, reverse saturation at $-I_0$, and the temperature dependence of $I_0 \propto n_i^2$.
- Explain how gate-controlled depletion width modulates channel conductivity in a J-FET, and identify the pinch-off mechanism that produces current saturation.
- Contrast the MOSFET operating principle (field-induced inversion layer) with the J-FET (gate-junction depletion), and explain pinch-off at $V_{DS} = V_{GS} - V_T$.

Optoelectronics (ch. 13)

- Compute the photon energy based on its frequency through $E = h\nu = hc/\lambda$.
- Contrast photon absorption in atoms (discrete lines, $h\nu = E_m - E_n$), metals (any $h\nu$, continuous empty states above E_F), semiconductors ($h\nu \geq E_g$ threshold), and insulators (E_g too large for visible photons).
- Apply the Beer-Lambert law $I(d) = I_0 e^{-\alpha d}$ to relate absorption coefficient, sample thickness, and transmitted intensity.

- Explain why thickness is so important when it comes to understanding light absorption in materials.
- Describe why even insulating materials can be opaque in the presence of defects (scattering)
- Explain how the band gap sets material color: photons with $h\nu \geq E_g$ are absorbed while lower-energy photons are transmitted, so the perceived color corresponds to the sub-gap portion of the visible spectrum.
- Describe how dopant-induced defect states inside a wide-gap insulator create selective absorption bands (e.g. Cr^{3+} in Al_2O_3 : green and violet absorbed $\rightarrow$ ruby appears red).
- Explain the ruby laser as a 3-level system: broadband pumping into $^4T_2/^4T_1$ bands, fast non-radiative decay to the metastable 2E level, population inversion, and stimulated emission at 694.3 nm.
- Distinguish direct and indirect band gap absorption: direct gaps (e.g. GaAs) produce a sharp onset with high α , while indirect gaps (e.g. Si, Ge) require phonon assistance and show a gradual, weaker onset.
- Explain why a step-function DOS in 2D (quantum wells) concentrates interband transition probability at the band edge compared to the $\sqrt{E}$ DOS of 3D bulk, favoring efficient light emission.
- Generalize the point before to understand why DOS relates to color even in metals (e.g., copper has a red-like hue)
- Recognize an LED: P-N junction where the forward bias injects electrons and holes into the junction region, radiative recombination emits photons with $h\nu \approx E_g$, and the semiconductor band gap determines the emission color.
- Recognize the operation of a solar cell: photon absorption at the junction creates electron-hole pairs, the built-in electric field separates carriers, and photocurrent flows through an external load converting light to electrical power.
- Explain how alloying semiconductors (e.g. $\text{In}_x\text{Ga}_{1-x}\text{N}$, $\text{Al}_x\text{Ga}_{1-x}\text{As}$) tunes the band gap and thus the emission/absorption wavelength across the visible and infrared spectrum.

Dielectric materials (ch. 10)

- Distinguish dielectrics from semiconductors based on bandgap magnitude (>2.5 eV) and explain why polarization effects dominate over charge transport.
- Connect the macroscopic relation $C = \epsilon_r C_0$ to the physical picture of bound charge accumulation at the dielectric surfaces inside a capacitor.
- Identify the three polarization mechanisms (electronic, ionic, orientational), explain their microscopic origins, and rank their characteristic response times ($\sim 10^{-15}$, $\sim 10^{-13}$, $\sim 10^{-9}$ – 10^{-6} s).
- Explain why ϵ_r decreases stepwise with increasing frequency as each polarization mechanism

freezes out: orientational above $\sim 10^9$ Hz, ionic above $\sim 10^{13}$ Hz, leaving only electronic at optical frequencies.

- Explain why the polarization response lags behind the applied field, leading to a complex dielectric constant $\varepsilon(\omega) = \varepsilon'(\omega) - i\varepsilon''(\omega)$.
- Interpret ε' as in-phase (energy storage) and ε'' as out-of-phase (energy dissipation), and relate the power absorbed per unit volume to $I = \omega \varepsilon_0 \varepsilon'' |E_0|^2$.
- Explain why piezoelectricity requires a non-centrosymmetric crystal: stress separates the positive and negative charge centers only when inversion symmetry is absent.
- Describe pyroelectricity as temperature-dependent spontaneous polarization in polar crystals, define the pyroelectric coefficient $p = \partial P_s / \partial T$, and identify the Curie temperature T_C where P_s vanishes.
- Explain ferroelectricity as reversible spontaneous polarization under applied E , interpret the hysteresis loop parameters (P_r , E_c , P_{sat}), and describe domain switching as the microscopic mechanism.

Magnetic materials (ch. 11)

- Distinguish diamagnetism ($\chi_m < 0$, $|\chi_m| \sim 10^{-5}$) from paramagnetism ($\chi_m > 0$, $\chi_m \sim 10^{-3}$), explaining diamagnetism as Lenz's law at the atomic scale (induced moments opposing the field) and paramagnetism as partial alignment of pre-existing permanent moments.
- Use the relation $B = \mu_0(1 + \chi_m)H$ and explain why only unpaired electrons contribute a net magnetic moment, connecting electron configuration (e.g. $\text{Fe}^{3+} 3d^5$ vs $\text{Zn}^{2+} 3d^{10}$) to paramagnetic or diamagnetic behavior.
- Describe ferromagnetism as spontaneous magnetization below the Curie temperature θ arising from exchange coupling between neighboring spins, and identify the phase transition to paramagnetic behavior above θ .
- Understand the meaning of the Curie-Weiss law $\chi_m = C/(T - \theta)$ to describe susceptibility above the Curie temperature and interpret the divergence of χ_m as $T \rightarrow \theta^+$ as the onset of spontaneous ordering.
- Explain why ferromagnets break into domains as an energy minimization process: a single domain has high magnetostatic (stray field) energy, antiparallel domains reduce it, and closure domains can eliminate it entirely, at the cost of domain wall energy.
- Describe domain walls as regions of gradual spin rotation, and explain how an applied field drives domain wall motion from equal domains ($H = 0$) through wall displacement (small H) to a single saturated domain (large H).
- Explain the microscopic origin of hysteresis: crystal defects pin domain walls in energy minima, and once a wall jumps past a pinning site the process is irreversible, producing path-dependent $B(H)$.

- Interpret the magnetic hysteresis loop parameters: remanence B_r (flux at $H = 0$), coercivity H_c (field to demagnetize), saturation B_s , and connect the loop area to energy dissipated per cycle via $E_d = \oint H dB$.
- Recognize the existence of ferromagnetic (parallel) and antiferromagnetic (checkerboard antiparallel) alignment for magnetization.
- Distinguish between soft magnets (high M_s , low H_c , narrow loop; transformers, motors) from hard magnets (high B_r , high H_c , wide loop; permanent magnets).